

Article

Integrating Digital Technologies in Sustainability Accounting and Reporting: Perceptions of Professional Cloud Computing Users

Monica Aureliana Petcu , Maria-Iulia Sobolevschi-David  and Stefania Cristina Curea 

Department of Financial and Economic Analysis and Valuation, Bucharest University of Economic Studies, 010374 Bucharest, Romania; maria.sobolevschi@cig.ase.ro (M.-I.S.-D.); stefania.curea@ase.ro (S.C.C.)

* Correspondence: monica.petcu@cig.ase.ro

Abstract: The configuration of economic development, through complexity, rhythm, and amplitude, imposed Industry 4.0 as a new industrial paradigm characterized by advanced digitalization and automation. Appropriate behavior of organizations from a sustainable perspective and the expansion of performance evaluations by considering economic–financial, social, and environmental aspects supposed the reconsideration of the role of accounting in reporting, advice on risk management, and determination of organizational policies. The application of digital technologies in sustainability accounting and reporting requires mapping the attributes of Industry 4.0 and the collection, processing, and reporting of financial and non-financial information. The purpose of this research consists of a correlative analysis of the impact of sustainable development constraints and Industry 4.0 on sustainability accounting and reporting. To collect information, we developed a semi-structured questionnaire, distributed to the members of The Romanian Chamber of Fiscal Consultants. The data have been processed using correlation and regression analysis. The odds of achieving a higher quality of information provided to stakeholders are greater for those interested in digital transformation, sustainable accounting and reporting digitization, while ensuring data protection and security. In the case of cloud computing users, the digitalization of sustainability accounting and reporting were identified as key factors in the process.

Keywords: digital technologies; cloud computing; sustainability accounting and reporting; stakeholders; practitioners



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1. Introduction

The dynamics of society's needs, their magnitude, and the compression of the response time to them have imposed structural industrial changes, embodied in Industry 4.0. This involves the IoT, real-time data collection, and predictive analytics using big data analytics, artificial intelligence, cloud manufacturing, and blockchain. The core principles of Industry 4.0 are interconnection and interoperability, information transparency (virtualization), decentralization and autonomous decisions, real-time capability, technical support and service orientation, and finally modularity [1]. Technological innovation is essential for the future of any successful business. By intensifying automation and digitizing processes, Industry 4.0 has a significant impact on organizations, improving efficiency and competitiveness through time savings, cost reduction, increased productivity, and system flexibility. Furthermore, adaptation to the new consumption patterns of customers is ensured. Capitalizing on the opportunities of these technologies, by customizing the manufacturing and delivery of goods and services to consumers, revolutionizes the market and creates value. This new industrial paradigm determines changes in business values, generating new business models, new strategic perspectives, proactive behavior, and a reconsideration of the social impact by focusing on the ability to influence, with a corresponding increase in the qualification level of employees.

The awareness of the need to assume responsible, pro-sustainable behavior has generated research streams, as well as the involvement of political decision-makers, the business environment, and the community. According to Brundtland Report sustainable development “meets the needs of the present without compromising the ability of future generations to meet their own needs” [2] (p. 15). In this context, the organization–society relationship must be assessed based on the legitimacy theory, the institutional theory, the resource dependence theory, and the stakeholders theory [3]. Ref. [4] defined legitimacy as a generalized perception and assumption that the actions of a company are desirable, proper, or appropriate within a social system, mediating the perception of interested parties on the need to obtain organizational legitimacy. Legitimacy theory states that businesses disclose social responsibility information to project an image of social responsibility and to legitimize the actions of their stakeholders. Complementarily, institutional theory defines schemes, rules, norms, and routines as authoritative guidelines for social behavior [5]. The high dynamics of economic development suppose dependence on resources and, consequently, particular actions of the entities. In this context, resource dependence theory studies the impact of procuring external resources on the behavior of the organization [6]. The inclusion of stakeholders in addressing the issue of economic entities facilitates the long-term framing of their development in agreement with the major trends of society as a whole [7]. Correspondingly, stakeholder theory was proposed [8], which involved a descriptive approach, to present and explain the characteristics and behaviors of entities; an instrumental approach, to determine the relationships that exist between stakeholders and the accomplishment of corporate objectives; and a normative approach, to establish “the moral or philosophical guidelines for the operation and management of the corporation” [9] (pp. 56–91).

Appropriate behavior of entities from a sustainable perspective and the expansion of performance evaluation by including not only economic–financial aspects but also social and environmental aspects require reconsideration of the role of accounting in the collection, processing, and communication of information. As sustainability becomes an increasingly relevant issue, it is important that accounting professionals are competent in how to handle and communicate such information. Having skills in collecting, measuring, and analyzing pertinent data and in the establishment of organizational policy regarding disclosure, accountants create frameworks that consider sustainability data, tailored to the entity’s needs to collect and compile social and environmental data. Accountants provide guidance on risk management, including the effects of voluntary reporting of sustainability issues for the entity. Furthermore, they are aware of the importance of understanding reporting environments and staying up to date on any changes that may affect the scope of reporting or statutory requirements. In the process of creating value both at the level of the organization and of society as a whole, sustainability accounting and reporting (SAR) was developed as a framework for defining sustainability variables based on a triple bottom line model, focused on the disclosure of non-financial information about a company’s performance to stakeholders. It is developed by Global Reporting Initiatives comprising universal standards of disclosures and management approaches of environmental, social, and governance goals. A sustainability report communicates these indicators, as well as an entity’s progress towards them, and the extent to which it creates value.

The integration of digital technologies in sustainable accounting and reporting requires mapping between the attributes of Industry 4.0 and data collecting, processing, and presenting financial and non-financial information. The rapid advancement of digital technology has profoundly influenced accounting practices. Core technologies such as cloud computing, big data analytics, artificial intelligence (AI), the Internet of Things (IoT), and blockchain have revolutionized traditional accounting processes, significantly enhancing efficiency, accuracy, and transparency. During the COVID-19 pandemic, cloud-based systems facilitated remote work and collaboration, substantially improving the efficiency of financial processes [10]. These systems also support operational processes, improve the scalability of accounting operations, strengthen data security measures [11], enhance

collaboration among stakeholders, and reduce IT infrastructure costs [12]. A notable example is the eXtensible Business Reporting Language (XBRL), which, facilitated by cloud platforms, simplifies the financial reporting process, ensuring regulatory compliance and improving efficiency in investment and business decision-making [13]. Big data analytics have been instrumental in enhancing audit quality by providing deeper insights into financial anomalies and detecting fraud [14]. The integration of big data into accounting practices allows for the processing of large volumes of financial data, pattern identification, and more informed decision-making. It supports predictive analytics, risk management, and performance evaluation [15]. Additionally, big data tools such as Hadoop and Spark offer robust frameworks for managing and analyzing complex datasets, thus improving decision-making processes [16]. Artificial intelligence (AI) in accounting varies from automating routine tasks to providing advanced analytical capabilities. Machine learning algorithms can predict financial trends and identify anomalies in financial statements, significantly reducing the risk of errors and fraud [17]. AI-powered chatbots and virtual assistants are also employed to handle customer inquiries, streamline audit processes, and provide real-time financial information [18]. The integration of the Internet of Things (IoT) in accounting enables continuous monitoring of financial transactions and asset management, providing more accurate and timely financial information [19]. This technology also supports inventory management by offering real-time data insights, reducing human labor, increasing precision and efficacy, and ensuring quality management [20]. By integrating the IoT with sustainability accounting systems, companies can achieve continuous monitoring and reporting of their sustainability performance. This real-time data collection helps organizations respond promptly to sustainability issues and make data-driven decisions to improve their environmental impact. IoT enhances the granularity and timeliness of sustainability reports, making them more relevant and useful for stakeholders [21]. Blockchain technology is recognized for its ability to provide an immutable and transparent ledger, reducing the risk of fraud and enhancing the reliability of financial statements [22]. Smart contracts facilitated by blockchain are being explored for automating compliance and regulatory reporting [23].

Digital technologies play a crucial role in facilitating integrated reporting by enabling the collection, analysis, and dissemination of diverse datasets. The synergy of these technologies creates a robust ecosystem for sustainability accounting and reporting. For instance, big data analytics aids in capturing environmental, social, and governance (ESG) metrics essential for IR [24]. AI and machine learning algorithms can analyze these indicators to provide insights into an organization's sustainability performance [25]. Cloud computing and big data analytics support comprehensive and timely integrated reporting, leading to better stakeholder engagement and decision-making [26]. AI and blockchain enhance the accuracy and credibility of integrated reports by ensuring data consistency and transparency [27]. Despite numerous benefits, the adoption of digital technologies in sustainability accounting and reporting (SAR) faces significant challenges, including concerns about data privacy, substantial investments in IT infrastructure, and the need for new skills among accounting professionals [28].

It is found that there is a lack of research in the field of the impact of digitalization on sustainability accounting and reporting. The purpose of this research consists of the correlative analysis of the impact of sustainable development and Industry 4.0 constraints on sustainability accounting and reporting. The research questions are as follows: does digital transformation ensure the conditions for the integration of digital technologies in sustainability accounting and reporting? (RQ1); is there an association between digital transformation, accounting digitization, data protection and security, digitization of sustainability reporting, and the quality of information provided to stakeholders? (RQ2); to what extent do digital transformation, accounting digitalization, data protection and security, and digitalization of sustainability reporting improve the quality of information provided to stakeholders? (RQ3); are there different perceptions in cloud computing users? (RQ4).

Considering the two-dimensionality of the approach, the contributions of the paper aim both at the aspects of the integration of digital technologies in sustainability accounting and reporting and at the identification of the necessary directions in reporting the performance of the entities to improve the quality of the information provided to stakeholders. First, the perception of accounting professionals regarding the use of digital technologies in general and cloud computing, in particular, is studied in the context of assessing the entity's behavior in terms of digital transformations. Second, the study finds applicability in substantiating entity policies to promote the increased use of digital technologies in sustainability accounting and reporting. By improving the quality of the information provided to stakeholders through the use of digital technologies, benchmarks are provided to reduce the conflict between the requirements of sustainable development and the behavior of economic operators circumscribed by the objective of profitability, increasing the interest of decision-makers in assuming a proactive role.

The following section contains the problem statement on transdisciplinary topics such as sustainability accounting and reporting, and cloud computing. The detailed description and process of research are presented in the Section 3. Information on the questionnaire, the sample, the areas of analysis in which the questionnaire items were included, and the methodology are provided. The empirical results and the discussions based on them are then highlighted, identifying the theoretical and practical implications. Section 6 summarizes the conclusions of the study and the future directions of research development.

2. Problem Statement

2.1. Sustainability Accounting and Reporting

As a tool for strategic management, sustainability accounting is an integrated system of financial measurement of economic, social, and environmental aspects that enables disclosure about the entity's achievements for sustainable development to important internal and external stakeholders, thereby supporting the entity's accountability [29]. Through the measurement of performance, sustainability accounting assesses the degree of implementation of sustainable development principles as the basis for settling the economic, social, and environmental responsibility of the entity and communicates results on achievements towards external stakeholders. The concept of sustainability accounting and reporting emerged from the need to make businesses more accountable and transparent about how their decisions affect the environment and society [30].

Sustainability accounting is a tool used by different stakeholder groups to measure, present, and communicate corporate sustainability performance. It also serves as a justification for the entity's environmental impact mandate and as a certification of the business's "moral maturity," attesting to its role in sustainable development. It is considered to be a new institutional component of the business's operations, connected to its role in sustainable development and the requirement to disclose its social, environmental, and economic performance. Sustainability accounting provides an "information signal" that managers send to entity statement recipients to eliminate knowledge asymmetry among stakeholders [31].

The users of information generated from the sustainability accounting system are grouped into external reporting users and internal reporting users [32,33]. To meet their need for information, measuring the entity's performance and sustainability reporting can be approached from an outside-inward perspective and an inside-outward perspective. The external approach is focused on meeting stakeholder expectations and external stakeholders' information requirements. The internal approach aims at the entity's business strategy and its effective implementation through sustainability. In this case, sustainability accounting is useful in the process of data collection and information communication to support the internal processes of implementing sustainable objectives. Managers' demands to communicate and place the company in the market and society have resulted in sustainability reporting, focusing on obtaining and providing information from problem-solving [34].

In the literature, a critical path of sustainability accounting development is identified [35]. Ref. [36] considers that it is difficult to precisely define what a sustainable entity would look like, and consequently the sustainability report, and also supports the primacy of the financial information, the conflict between it, and other bottom lines. There is a suspicion regarding entity motives for sustainability reporting [37], with entities trying to create a favorable image by becoming involved in social and environmental issues. The critical stream of accounting research mentions entities' environmental and social misconduct and the perceived inability of companies to contribute to sustainable development [38]. Customizing the components of the traditional accounting and financial reporting framework, sustainability accounting and reporting includes objectives of the sustainability accounting framework; principles that underpin the application of the framework; data capture tools, accounting records, and measurement techniques; reports used to present information to stakeholders; and qualitative attributes of information reported using the framework [39]. The Corporate Sustainability Reporting Directive (CSRD) was issued in 2022 as a fundamental framework, acting as a catalyst for openness, accountability, and beneficial environmental and social consequences in the corporate world, in response to the pressing need for organizations to embrace sustainable practices. Although previous research made valuable contributions in terms of the Non-Financial Reporting Directive's (NFRD) impact on business, the new directive requires all large companies, regardless of the type of legal structure under which they operate, to report on their sustainability impacts, beginning 2026. Complementarily, considering the intrinsic characteristics of the subject matter, a sustainability report includes both quantitative aspects, which can be verified, as well as qualitative aspects, which involve subjective reasoning and cannot be tested; it is not possible to guarantee highly reliable verification of the information [40].

The main challenge for accounting innovation is to make sure that measurements truly take into account science-based targets as well; the information created and indicators applied are linked to broader sustainability issues and societal goals of sustainable development (as addressed by the UN Sustainable Development Goals [41]); and the environmental issues that are most pertinent to the industry and situation are those in question [38].

To ensure the relevance of the reported information, the qualitative characteristics of information from a sustainability accounting system should be as follows: accurate: the data gathered from SARs must be detailed and accurate enough for stakeholders to evaluate the performance of the reporting entities; balanced: in order to allow for a rational evaluation of overall performance, the data from the sustainability accounting system must fairly represent both the reporting organization's good and bad performance; clear: the entity must provide information in a way that stakeholders using it can comprehend and access; comparable: the data from the SAR system must be displayed so that interested parties can assess how the entity's performance has changed over time; reliable: the organization must collect, record, compile, evaluate, and report data and procedures used in the report's creation in a manner that allows scrutiny and determines the information's relevance and quality; and timely: the information should be timely in order for stakeholders to have enough time to make informed decisions [31].

Sustainable development is characterized by multi-dimensional, multi-disciplinary, political and scientific, dynamic, and complex non-linear relationships. Accountability in this context aims at the potential of accounting as a transformative and emancipatory force, rather than a partial description and disclosure, of interest to corporations, governments, and regulators [42].

Complementarily, it is necessary that this information be collected, processed, and reported using viable methods and tools. Thus, the quality and efficiency of the process require the integration of digital technologies in sustainability accounting and reporting.

2.2. Cloud Computing

Cloud computing represents a modern paradigm for delivering computing services over the Internet, providing access to resources such as servers, storage, databases, net-

works, and software. This technology enables organizations to benefit from rapid innovation, flexibility, and economies of scale without requiring significant investments in physical infrastructure [43]. Cloud computing is classified into three architectural models: Software-as-a-Service (SaaS), Platform-as-a-Service (PaaS), and Infrastructure-as-a-Service (IaaS), each with its own characteristics and benefits [44].

SaaS is a software distribution model in which applications are hosted by a provider and accessed by users via the Internet. Updates are managed by the provider, ensuring that users always have access to the latest version of the software. This eliminates the need for software installation and maintenance on local devices, providing rapid access to complex and continuously updated applications such as Google Workspace, Microsoft Office 365, and Salesforce.

PaaS offers a complete platform and environment for developing, testing, and deploying applications. This platform allows developers to focus on writing code and application logic without worrying about managing the underlying infrastructure. The model provides support for collaboration, automatic scalability, and integrated development tools. Examples of PaaS services include Google App Engine and Microsoft Azure PaaS.

IaaS provides virtualized computing resources such as servers, storage, and networks that users can rent and manage according to their needs. This offers maximum flexibility and control over IT infrastructure, while also reducing the costs associated with purchasing and maintaining physical hardware. IaaS models include Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP).

Implementing cloud computing requires careful planning and change management, including the following steps:

- Needs assessment: identifying the processes that can benefit the most from migrating to the cloud;
- Choosing a provider: selecting a cloud service provider that offers solutions compliant with organizational requirements and industry regulations;
- Data migration: ensuring a secure and efficient transition of data to the cloud;
- Staff training: training the team to use new technologies and maximize the benefits brought by cloud computing.

Implementing cloud computing offers significant advantages for organizations of all sizes and industries:

- Scalability: scalability is one of the defining characteristics of cloud computing, allowing for the rapid adjustment of IT resources to meet dynamic user demands. There are two main types of scalability:
- Vertical scalability (scaling up or down): this involves adding or removing resources (e.g., CPU, RAM) on an existing server to manage variations in resource demand;
- Horizontal scalability (scaling out or in): this involves adding or removing instances of servers to distribute the load and improve application performance and availability [45];
- Cost reduction: the pay-as-you-go model reduces the need for large initial investments in IT infrastructure and allows organizations to pay only for the resources used [46];
- Flexibility: organizations can quickly scale IT resources to meet fluctuating demands, enhancing their ability to adapt to market changes [44];
- Global accessibility: resources and applications can be accessed from anywhere, facilitating remote work and global collaboration;
- Automatic updates: cloud service providers handle system updates and maintenance, ensuring constant access to the latest technologies;
- Advanced security: most cloud service providers offer robust security solutions, including encryption, multi-factor authentication, and strict access policies [47];
- Business continuity: Cloud computing offers robust backup and disaster recovery solutions, ensuring the availability of data and applications in any situation. Data are stored in multiple geographic locations, providing redundancy and reducing the risk of data loss [43].

Despite the numerous advantages of cloud computing, its use also presents a series of challenges that need to be carefully managed:

- Data security and privacy: Transferring data to the cloud can involve security and privacy risks. Organizations need to implement adequate data protection measures and collaborate with cloud providers who comply with security standards [43];
- Vendor lock-in: Complete migration to the cloud can lead to excessive dependence on a single provider. Organizations need to carefully evaluate the terms and conditions of the services provided to avoid vendor lock-in situations [48];
- Transfer costs: Migrations between different cloud providers or between on-premises infrastructure and the cloud can involve significant costs and complexity. Detailed planning and evaluation of these migrations are essential to minimize financial and operational impact [46];
- Service availability: Cloud provider service interruptions, although rare, can affect the accessibility of data and applications. Organizations need to have business continuity plans and backup solutions [45].

Cloud computing has become an essential element in the digital transformation of accounting practices, offering flexible, scalable, and efficient solutions for financial management and reporting. Integrating cloud technologies into accounting facilitates process automation, improves data access, and increases the transparency and accuracy of financial information. A study by PwC [49] shows that 78% of organizations that adopted cloud accounting solutions reported a significant improvement in operational efficiency. Additionally, a Deloitte report [50] highlights that the use of cloud technologies in accounting led to a 30% reduction in financial reporting errors.

3. Materials and Methods

To analyze the impact of integrating digital technologies in sustainability accounting and reporting, a survey based on a semi-structured questionnaire (Appendix ??) was carried out. The questionnaire was distributed to the members of the Chamber of Fiscal Consultants from Romania between 4 March 2024 and 15 April 2024. In the structure of the questionnaire, the first three items are intended to present the characteristics of the respondents' entities (age; type: micro-entity, small entity, medium/large entity, association/foundation, public institution; digital technology used: cloud computing, big data analysis, artificial intelligence, IoT, blockchain, others); the fourth item concerns the quality of member of professional bodies (Romanian Chamber of Fiscal Consultants—RCFC, Romanian Chamber of Certified and Chartered Accountants—RCCCA, Romanian Chamber of Financial Auditors—RCFA, other); and the following thirty-seven items aim to characterize the extent to which the application of digital technologies in accounting and reporting ensures an improvement in the quality of information provided to interested parties, through the explanatory variables. For this last category of items, the default answers, converted to a Likert scale from 1 to 5, were as follows: to a very small extent (1), to a small extent (2), neutral (3), to a great extent (4), and to a very large extent (5). These items were concatenated into four relevant areas, digital transformation, accounting digitalization, data protection and security, and digitalization of sustainability reporting, to evaluate the improvement of the quality of information provided to interested parties through the application of digital technologies, defining the analysis variables.

The digital transformation is evaluated by the existence of reluctance to the change implied by the use of digital technologies; the extent to which the organization promotes the digitization of services previously provided by other means and uses digital solutions for business problems; the skills and abilities required to use the technologies possessed by the staff; the existence of leadership that promotes and supports the digital transformation;; and the financial resources necessary to implement the digital technologies available to the entity.

Accounting digitalization is assessed based on the following aspects: the extent to which the use of digital technology facilitates the collection and analysis of reliable sus-

tainability data, in the context of the complexity related to the involvement of several interested parties and the need for solid data management systems; the extent to which the entity ensures the automation of routine, repetitive, and structured sustainable accounting operations, respectively, of non-repetitive sustainable accounting processes; the extent to which the organization provides the digital workspace, the necessary infrastructure, and the outsourcing of accounting services at the employees' homes; the degree of adaptation and integration of digital technologies with accounting and integrated reporting standards; the degree to which digital technologies allow the tracking and accessing of sustainable accounting in real-time; the digital expertise that the accountant must have to measure, analyze, and correctly interpret financial and non-financial data; and the extent to which the use of digital technology enables accounting professionals to collaborate with cross-functional teams to set performance goals, monitor progress, and provide assurance on the reliability of sustainability information.

Data protection and security are evaluated by the entity's interest in adopting shared data access policies and practices, data security, restriction in the creation and deletion of operational-functional structures, and data retention.

In order to evaluate the digitalization of sustainability reporting, we investigated the extent to which the application of digital technologies ensures improvements in the quality of financial and non-financial reporting; real-time access to data by interested parties; faster adoption of decisions by investors, employees, customers, and regulatory bodies; the management of risks and trust building with stakeholders; standardization of reporting practices, in the absence of universally accepted reporting standards for ESG metrics; increased transparency, accountability, and stakeholder engagement by effectively communicating sustainability information through the use of digital technology; improved access to capital by attracting responsible investors; sustainability concerns focused on internal processes, respectively, and on the entire lifecycle of the product/service; improved operational efficiency and reduced costs, driving innovation by identifying opportunities for sustainable practices along their value chains.

In total, 347 people answered the questionnaire, with all answers being valid. The size of the sample studied is relevant, being superior to that determined by applying Cochran's Formula for Sample Size, considered appropriate for populations that are heterogeneous in nature, at a 95% confidence level (which gives Z values of 1.96, per the normal tables), with a $\pm 5\%$ margin of error. Internal consistency analysis was conducted using Cronbach's alpha, which indicates how closely related a set of items are as a group. The Cronbach's alpha coefficients are 0.7995 for digital transformation, 0.9347 for accounting digitalization, 0.8827 for data protection and security, and 0.9731 for the digitalization of sustainability reporting. All these values exceed the 0.70 thresholds considered acceptable in social science research, demonstrating high internal consistency among the variables. To evaluate the data used in exploring and formulating conclusions, we employed descriptive statistics (median, maximum, minimum). These indicators, calculated for both the entire sample and the considered subsamples, enable us to characterize respondents' perceptions of the variables of interest and identify differences between them. The strength and direction of the linear association between variables were measured using the Spearman coefficient.

The assessment of the improvement of the quality of information provided to interested parties through the application of digital technologies was made based on a regression analysis, using the ordinal logit model (OLM) (Table 1).

To verify the validity of the model, we checked its robustness by considering some subsamples: cloud computing users ($n = 236$), the opinion of those who are members of several professional bodies ($n = 296$), and cloud users ($n = 202$). To determine how well the model fits the data, the Likelihood Ratio (LR) statistic and the measures of association were examined.

Table 1. The variables of the model.

Variables	Domains	Abbreviations	Items *
Independent variable	digital transformation	DT	5–10
	accounting digitalization	AD	11–22
	data protection and security	DPS	23–26
	digitalization of sustainability reporting	DSR	27–40
Dependent variable	information quality	IQ	41

* question number in the questionnaire.

4. Empirical Findings

The analysis of the data highlights a favorable perception of the practitioners regarding the impact of the integration of digital technologies in sustainability accounting and reporting, regardless of the age and the type of entity. At the level of the studied sample, the main digital technologies used are cloud computing ($n = 236$), big data ($n = 28$), artificial intelligence ($n = 21$), IoT ($n = 7$), and blockchain ($n = 6$). The perception is similar both among RCFC members and in the case of those who are also members of other professional bodies, with higher median values regarding the digitalization of accounting in the case of respondents who are also members of RCCCA, respectively, and RCFA. The assigned values are between a maximum of 5 and a minimum of 1 in all cases, except digital transformation, for which a more intense involvement of the entity is perceived. Regarding cloud computing users, the opinions are similar; they hold the majority of the sample, without any deviations depending on the membership of various professional bodies (Table 2).

Table 2. Descriptive statistics of variables.

Variable	DT	AD	DPS	DSR	IQ	DT	AD	DPS	DSR	IQ
	RCFC members					Members of other professional bodies				
Median	3.67	3.58	3.75	3.79	4	3.67	3.79	3.75	3.71	4
Maximum	5	5	5	5	5	5	5	5	5	5
Minimum	1.17	1	1	1	1	1.17	1	1	1	1
	Cloud computing					Cloud computing				
Median	3.67	3.75	4	3.78	4	3.67	3.75	4	3.78	4
Maximum	5	5	5	5	5	5	5	5	5	5
Minimum	1.17	1.25	1.5	1.14	1	1.17	1.25	1.5	1.14	1

Source: Elaborated by the authors in the EViews programme.

The analysis of the correlation between the variables (Table 3) points out the associations between them (RQ2).

Table 3. Correlations between variables: RCFC members.

	RCFC Members					Cloud Computing Users				
	DT	AD	DPS	DSR	IQ	DT	AD	DPS	DSR	IQ
DT	1					1				
AD	0.7593 *	1				0.7321 *	1			
DPS	0.6699 *	0.7226 *	1			0.6477 *	0.7568 *	1		
DSR	0.5315 *	0.6282 *	0.6170 *	1		0.5428 *	0.6541 *	0.6605 *	1	
IQ	0.5587 *	0.6282 *	0.5924 *	0.7618 *	1	0.5607 *	0.6492 *	0.5876 *	0.7016 *	1

Note: * $p < 0.001$. Source: Elaborated by the authors in the EViews programme.

At the level of the entire sample, the existence of positive correlations, statistically significant between the variables of the model, can be observed. On the subject of the strength of the association, there is a strong correlation between information quality for

stakeholders ensured by the use of digital technology in sustainable accounting and reporting, and accounting digitalization, respectively, as well as digitalization of sustainability reporting. There is a moderate correlation between information quality for stakeholders and digital transformation, and data protection and security. Furthermore, accounting digitalization is strongly associated with digital transformation, data protection and security, and digitalization of sustainability reporting. There are similar associations in the case of digitalization of sustainability reporting and data protection and security, as well as between digital transformation and data protection and security. The strongest association is between digitalization of sustainability reporting and information quality for stakeholders, and the lowest, moderate correlation is between digital transformation and digitalization of sustainability reporting. The opinion of cloud computing users, who hold the main weight in the sample (68.01%), expresses the meaning and intensity of the strength of the correlation at the level of the analyzed subsamples. The strongest correlation is between accounting digitalization and data protection and security, maintaining at the lowest level the moderate correlation between digital transformation and digitalization of sustainability reporting.

Associations of the same meaning, statistical significance, and intensity are also found in the case of the sample of respondents who are members of both the RCFC and other professional bodies in the field (Table 4).

Table 4. Correlations between variables: RCFC, RCCCA, RCFA members.

	RCFC, RCCCA, RCFA Members					Cloud Computing Users				
	DT	AD	DPS	DSR	IQ	DT	AD	DPS	DSR	IQ
DT	1					1				
AD	0.7621 *	1				0.7395 *	1			
DPS	0.6575 *	0.7088 *	1			0.6175 *	0.7458 *	1		
DSR	0.4941 *	0.6245 *	0.6212 *	1		0.5 *	0.6617 *	0.6463 *	1	
IQ	0.5290 *	0.6126 *	0.6013 *	0.7331 *	1	0.5341 *	0.6354 *	0.5792 *	0.6788 *	1

Note: * $p < 0.001$. Source: Elaborated by the authors in the EViews programme.

The correlation between data protection and security and information quality for stakeholders is strong in this case, without significant changes. The strongest association is between accounting digitalization and digital transformation, and the lowest, moderate correlation is between digital transformation and digitalization of sustainability reporting. In the case of cloud computing users, the strongest correlation exists between accounting digitalization and data protection and security, and the lowest correlation, moderate in intensity, is also between digital transformation and digitalization of sustainability reporting.

The assessment of determinisms in relationship information quality for stakeholders ensured by the use of digital technology in sustainable accounting and reporting and the domains considered (DT, AD, DPS, DSR) is carried out using regression analysis. Estimates for the regression equations in the case of RCFC members, in total (model 1), respectively, and cloud computing users (model 2) are presented in Table 5.

The odds ratio of 1.5342 suggests that the odds of achieving higher information quality are 53.42% greater in the case of those who increase with one-unit digital transformation, compared to those who have a one-unit decrease. The 0.5645 coefficient reveals that when accounting digitization increases by one unit, the log odds of achieving higher information quality for stakeholders increase by approximately 0.5645 units, with other variables being constant. The odds ratio is 1.7586, highlighting that the odds of achieving a higher information quality are 75.86% higher for those with a one-unit increase in accounting digitalization compared to those who have a one-unit decrease, under the conditions of keeping the other variables constant. With a coefficient of 0.3982, there is a reference that an increase of one unit of data protection and security is associated with an increase in the log odds of improving information quality by 0.3982. The odds ratio is 1.4892, suggesting that the odds of achieving higher stakeholder information quality are 48.92% greater in the case

of those concerned with increasing data protection and security, compared to those who have a one-unit decrease in this interest. Furthermore, as the digitalization of sustainability reporting increases, it is more likely that information quality will improve.

Table 5. Results of the regression: model 1 and model 2.

Dependent Variable: IQ				
Independent Variable	Model 1: RCFC Members		Model 2: Cloud Computing Users	
	Coefficient	Prob.	Coefficient	Prob.
DT	0.427993	0.0557	0.334232	0.2204
AD	0.564508	0.0425	1.565595	0.0001
DPS	0.398232	0.0478	0.243211	0.3805
DSR	2.225963	0.0000	1.999633	0.0000
Pseudo R-squared	0.412501		0.416523	
LR statistic	376.5074		241.0334	
Prob(LR statistic)	0.0000		0.000	
odds ratio				
DT	1.534175		1.396867	
AD	1.758583		4.785522	
DPS	1.48919		1.275338	
DSR	9.262401		7.386342	

The low values of the p -value reveal a high degree of confidence regarding the existence of a relationship between each independent variable and information quality.

The pseudo-R-squared value suggests that approximately 41.25% of the variability of information quality for stakeholders is explained by the model. The Likelihood Ratio (LR) statistic tests the goodness of fit of the model, with a high value of 376.51 indicating that the model with predictors offers a significantly better fit to the data. A value of 0.0000 of the prob (LR statistic) shows that the improvement in the model's fit with the digital transformation, accounting digitalization, data protection and security, and digitalization of sustainability reporting predictors is statistically significant, as a whole, having a significant effect on the improvement of stakeholder data quality.

It is thus demonstrated that digital transformation ensures the conditions for the integration of digital technologies in sustainability accounting and reporting (RQ1) and it is determined to what extent digital transformation, accounting digitalization, data protection and security, and digitalization of sustainability reporting improve the quality of information provided to stakeholders (RQ3).

In the case of RCFC members, users of cloud computing (Model 2), digital transformation, and also data protection and security become statistically insignificant, relevant in determining the quality of information provided to stakeholders, being accounting digitalization and digitalization of sustainability reporting. There is an increase in the probability that information quality for stakeholders will improve as the degree of digitization of accounting increases. Different perceptions are found in cloud computing users (RQ4).

To verify the validity of the model, its robustness was checked by considering some subsamples: RCFC members who are also members of other professional bodies such as RCCCA and RCFA, in total (Table 6, model 3), respectively, and users of cloud computing (Table 6, model 4).

Similar aspects are revealed in the case of the analyzed subsamples, specifying the fact that digital transformation becomes a statistically insignificant variable in the case of respondents who are members of various professional bodies in the field (RQ1). The odds ratio values reveal strong associations between variables, in the sense that the concern for the digitization of accounting and the digitization of sustainability reporting raise the odds of improvements in information quality for stakeholders, compared to the absence of these interests (RQ3). In model 3, there is a similar association in the case of data protection and

security and information quality (RQ3). Cloud computing users and members of several professional bodies have similar perceptions to those of RCFC-only members (RQ4).

Table 6. Results of the regression: model 3 and model 4.

Dependent Variable: IQ				
Independent Variable	Model 3: RCFC, RCCCA, RCFA Members		Model 4: Cloud Computing Users	
	Coefficient	Prob.	Coefficient	Prob.
DT	0.390571	0.1267	0.330715	0.2782
AD	0.610157	0.0556	1.385058	0.0020
DPS	0.445697	0.0460	0.406535	0.1827
DSR	2.120626	0.0000	1.896231	0.0000
Pseudo R-squared	0.402198		0.397528	
LR statistic	302.3835		192.9688	
Prob(LR statistic)	0		0	
odds ratio				
DT	1.477824		1.391963	
AD	1.84072		3.995059	
DPS	1.561578		1.501606	
DSR	8.336351		6.660742	

5. Discussion

The research provides valuable insights regarding the opportunities and challenges of integrating digital technologies in sustainability accounting and reporting.

Practical implications

The usefulness of digital technologies in sustainability accounting and reporting can be analyzed by considering their core principles (Table 7):

Table 7. The relevance of principles of Industry 4.0 in sustainability accounting and reporting.

Industry 4.0 Principles	Utility in Sustainability Accounting and Reporting
agility	real-time feedback and monitoring are ensured, which allow for great flexibility and agility, as well as rapid decision-making appropriate to different situations by different categories of stakeholders
interconnection	provides a coherent overview of ESG aspects, based on the concatenation of the exact knowledge of the problems specific to each point of the chain, on highlighting, monitoring, and reporting as a whole, within the interconnected chain, specific to the multidimensionality of sustainable development problems
information transparency (virtualization)	the transparency offered by Industry 4.0 technology provides accountants with complete data to inform stakeholders. Large amounts of data can be collected from various parts of the value chain and processed, determining specific indicators and evaluating ESG performance, to find correlations and, eventually, to build models in order to improve future behaviors. Not just information is available in the cloud but all objects, processes, and systems are transformed into virtual objects, facilitating process optimization
decentralization and autonomous decisions	these involve placing artificial intelligence in every tool and allow independence during decision-making based on information ESG available in the cloud. The ability of cyber systems to make decisions and perform their tasks as autonomously as possible ensures complete, accurate, and timely information
technical support and service orientation	the technological facility of systems to assist accountants in carrying out complex, difficult tasks, in making decisions, and in solving problems improves the process of collecting, processing, and reporting multidimensional, complex aspects

Digitization and automation of physical accounting processes save time, replace repetitive accounting operations, increase the degree of predictability, and enable the possibility of integrating some complex aspects into analyses and reports. The perspective of the ontology of interconnected events, namely the fact that parties do not exist in isolation [51], in the approach to sustainable development issues reveals the need to integrate digital technologies into sustainability accounting and reporting.

In the process of digitizing accounting, difficulties were identified: access to the Revenue Administration in an electronic environment, insufficient staff skills, lack of communication regarding the digitalization process, higher cost of services and work-load, constant technical problems in the use of e-document applications, and failure to adapt the IT system to legislative changes in real-time [52].

At the level of the studied sample, the integration of digital technologies is desirable, with the manifestation to a small extent of reluctance in this regard. Also, entities promote, to a great extent, the digitization of services previously provided by other means, as well as the use of digital solutions for business problems. Digitization not only alters how people work but it also quickens the pace of change that businesses must adapt to. The consequences lead to three key requirements that must be handled for success: new organizational capacities, new leadership styles, and new skills and competencies [53]. Regarding the conditions necessary for the automation and digitization of processes, the respondents attach greater importance to the need for the staff to have the skills and abilities necessary to use technologies to a large extent than to the existing leadership to promote and support the digital transformation and financial resources. Organizations experiencing digital transformation must consider the necessary skills and capabilities of their staff as a precondition for implementation and as an outcome of the change [54]. As organizations reunite and evolve digital technologies, it is important to comprehend the coevolution between the trajectories of connected digital technologies, firm's innovation routines, and skills development. Since digital technologies shift the codification of knowledge for productive, creative processes, organizations need skills to develop, acquire knowledge, and adjust to expanding digital technologies [55].

The procedures of accounting and workflow are evolving due to digital transformation. The accounting industry is becoming less and less focused on recording operations in favor of supporting management and decision-making. Accounting activities have evolved from data processing to data analysis [56]. This might result in the elimination of entry-level accounting jobs while also creating new responsibilities for accountants [57]. One of the most challenging changes in accountants' activity is to develop feedforward controls, to the same extent as feedback controls, which involve the monitoring of both internal and external variables [58]. Digital technology, complementary to process automation, also generates large databases. To contribute to value creation and the sustainability of companies, accountants will have to develop a higher level of transversal skills, such as technological and analytical ones.

Evaluating the integration of digital technologies in accounting reveals the fact that under the conditions of the complexities related to the involvement of several interested parties and the need for solid data management systems, the use of digital technology greatly facilitates the collection and analysis of reliable sustainability data. This requires accountants to have the appropriate digital expertise to measure, analyze, and correctly interpret financial and non-financial data, as well as professional accountants working with cross-functional teams to set performance targets, monitor progress, and provide assurance regarding the reliability of information regarding sustainability. Also, to a large extent, entities ensure the automation of routine, repetitive, and structured sustainable accounting operations, with an average level of interest in the automation of non-repetitive sustainable accounting processes by those related to relations with interested parties. Advanced algorithms and machine learning techniques are used in accounting to automate repetitive operations like data input, reconciliation, and financial reporting [56]. Accounting

practitioners believe that digital technologies must be adapted and integrated to a greater extent with accounting and integrated reporting standards.

The protection and security of data and the entire system is a key concern when using digital technologies. Ransomware attacks, malware, and the breaking of IT&C infrastructures have become a growing trend, increasing the risk that the wrong agents disrupt activity and have access to non-public data. In the literature, the main recommendations regarding information security are aimed towards the implementation of defense techniques such as endpoint protection, network segmentation, and access permission, ensuring compliance with relevant regulations [59,60] and ensuring that data security rests with cloud users. The respondents consider that entities adopt, to a large extent, policies and practices of shared data access, data security, restriction in the creation and deletion of operational–functional structures, and data disposal and retention to protect the data.

The respondents consider that the application of digital technologies enables, to a large extent, improvements in the quality of financial and non-financial reporting, real-time access of interested parties to data, and, consequently, faster adoption of decisions by investors, employees, customers, and regulatory bodies. Automation and digitization of the reporting process enable risk management and trust building with stakeholders. In the absence of universally accepted reporting standards for ESG metrics, the use of digital technologies ensures a standardization of reporting practices. At the same time, by effectively communicating sustainability information through the use of digital technology, transparency, accountability, and stakeholder involvement can be increased. Sustainable performance determines the behavior of investors in the case of similar financial and operational performances [61]. By integrating digital technology into accounting and sustainability reporting, entities can improve the operational efficiency of internal processes and the entire product/service lifecycle, reduce costs, and drive innovation by identifying opportunities for sustainable practices throughout their value chains. The dissemination of relevant information for the sustainable behavior of the organization improves access to capital by attracting responsible investors.

Nine in ten companies think the cloud is essential for growth, especially used in combination with other technologies such as artificial intelligence (AI), the Internet of Things (IoT), and analytics [50]. The cloud is a large pool of easily usable and accessible virtualized resources (hardware, development platforms, and/or services), which can be dynamically reconfigured to adjust to a variable load (scale) [48]. The main characteristics of cloud computing, relevant for use in sustainability accounting and reporting, are as follows:

- User friendliness/ease of use [48,62]: the ease of using cloud computing justifies its integration in most organizations;
- Internet-centric [48]: the requirements regarding access and the quality of the Internet do not constitute a problem in Romania, which benefits from a high average download and upload speed at the national level both for fixed connections and for mobile ones;
- Variety of resources [48] and business models [62]: Considering the diversity of solutions offered by cloud accounting providers, to improve the entity's ability to adapt to market changes and improve competitiveness, the choice of the specific service variant needs to be based on the business's specific characteristics. Data quality and storage, specific information and the best timing for the transfer, and the structure of the report requested by various interested parties are considered. Also, the cost of cloud accounting can be higher, including the cost of implementation, customization and staff training, and the maintenance fee [63];
- Automatic adaptation [48]: cloud solutions are quickly adapted to business needs, speeding up implementation in sustainability accounting and reporting processes without prior massive investments in infrastructure;
- Scalability [48]: Scalability in the cloud is essential to cope with demand growth and to ensure the application's performance and availability. The cloud services used in sustainability accounting and reporting can be adapted according to specific

needs. Entities can scale infrastructure efficiently, dynamically aligning to changes and business priorities, minimizing resource consumption. Choosing the right scalability solutions depends on the requirements and structure of the application or cloud service;

- Pay per use [48]: entities pay exclusively for the resources and services used;
- SLA's infrastructure [48]: through many options for finishing and choosing materials, accurate prototypes, and superior models at a comparatively quick pace, the necessary infrastructure in sustainability accounting and reporting is provided;
- Service-oriented [62,64]: Cloud computing is a technology which provides access to databases, servers, and software directly through the Internet. The cloud involves infrastructure-as-a-service, ensuring a huge capacity of processing, storage, and networks; platform-as-a-service, which mediates the connection between the hardware and application; and software-as-a-service, which replaces the applications running on PC, through the pay-per-use pattern [62]. It is effective in sustainability accounting and reporting, both through the costs it involves and through its agility;
- Loose coupling [62]/virtualization [48]: Loose coupling is the fundamental technical element of cloud computing; through virtualization or other technologies, the infrastructures are separated into logical or physical. Since each component can be made to use standardized interfaces or APIs to connect with other components, loose coupling can facilitate the integration of various systems or components. Because it lessens the effect of changes to one component on the other components, loose coupling can increase maintainability. Thus, loose coupling is effective in sustainability accounting and reporting because it reduces the risk that a change made in a database or a user interface will create unforeseen changes in other parts;
- Heavily fault-tolerant [62].
- Resource optimization [48]: the other mentioned characteristics support the ability to use the cloud to optimize the resources of organizations in conditions of superior efficiency.

The use of cloud computing in sustainability accounting and reporting aims at creating a framework for sustainability-related disclosures, methods, techniques, indicators, and/or metrics for measuring sustainability and identifying the determinants and implications of sustainability accounting and interrelationships between traditional financial reporting and sustainability accounting. Considering its characteristics, cloud computing is used in accounting activity in the customized accounting monograph, the automatic generation of accounting notes, the possibility of reporting costs and revenues on cost and profit centers, the automatic checkup of the correlation between management and financial accounting, the preparation of accounting reports, including based on different financial reporting standards (IFRS, GAAP), and the calculation of indicators [65]. The integration of cloud computing in sustainability accounting and reporting, through cost-effectiveness, accessibility, accuracy, and real-time interaction, ensures an improvement in the quality of the information offered to various categories of stakeholders. The new opportunities of using the cloud consist of increasing the speed to adoption and driving business process transformation [66], which is appropriate in sustainability accounting and reporting.

The main cloud accounting risks are structured in general risks associated with cloud accounting—hardware and software compatibility, Internet stability and availability, cloud accounting server stability and availability, data security, data maintenance, and backups—and specific risks to cloud accounting systems and services—financial statement reliability, legal compliance, location of data, and ownership of data [67]. Being a technology aimed at data storage and management, in the context of the increase in cyber threats, the cloud implies advanced protection, an additional level of security, with advanced technologies for identifying and neutralizing attacks. Data stored in the cloud are also encrypted, protecting unauthorized access. KPMG specialists have identified the main security-related challenges of cloud computing integration: data loss and privacy risks, general security risks, risk of intellectual property theft, legal and regulatory compliance, and system availability and

business continuity risks [66]. There is a risk that certain cloud users may accidentally gain access to certain data through certain deficiencies in the cloud system.

Theoretical implications

Research in innovation in the field of sustainable accounting aims at the creation of methods, software, guidelines, or standards. Considering the complex nature of sustainability challenges, the integration of digital technologies enables the identification of certain determinants in the economic–social–environmental relationship. The digitization of sustainability accounting and reporting offers the necessary tools for the development of accounting methods and for the correlation of measured indicators with a target of sustainable development at the societal level. In this way, sustainability management control systems are ensured, whose effectiveness is evaluated based on the proposed measures and the degree to which they lead to the achievement of the established objectives.

The transdisciplinary nature of accounting sustainability requires teams of professional with different expertise [68]. Complementarily, Industry 4.0 enhances integration between disparate systems, networks, people, and organizations for efficient decision-making [69].

In order to comply with present reporting procedures and satisfy stakeholder expectations, more standardized measurements and frameworks of various sizes are required. These must be incorporated into the accounting regulatory framework, which also includes professional associations for accounting as a forum for information exchange [70].

The integration of Industry 4.0 with a preponderance in capital-intensive industries requires one to focus research on environmentally sensitive sectors to ensure a significant impact of the disclosure and an increased efficiency of decisions.

6. Conclusions

The increase in concerns regarding sustainability, with holistic interdependencies, which involves a variety of interested parties, requires a change in the structure and quality of the information provided by the entities. It is thus necessary to create a suitable framework for sustainability accounting and reporting. The development of digital technologies provides the necessary infrastructure for the collection, processing, analysis, and reporting of a wide range of data corresponding to the information requirements of different categories of stakeholders. The integration of the constraints of sustainable development circumscribed by the use of digital technologies in the strategies of entities implies a cultural change in the way activities are carried out based on the development of an analytical mentality to have the ability to understand, process, and analyze a complex and large volume of information.

This research underscores the need for an integrated framework encompassing advanced Industry 4.0 technologies and sustainable development practices to ensure efficient and transparent accounting and reporting. Through its transdisciplinary character, the study has made several significant contributions to the field. To assess the entity's behavior regarding digital transformations, the perception of accounting professionals in terms of the use of digital technologies in general and cloud computing in particular is investigated. In addition, the research suggests that entity policies that encourage greater use of digital technologies in sustainability accounting and reporting are likely to be more effective in reducing information asymmetry across stakeholders. Improving the quality of information that stakeholders receive in the context of using digital technologies creates benchmarks that reduce the tension between the demands of sustainable development and the decisions of companies related to profit margins. Complementarily, the decision-makers adopt more active behavior to ensure a framework that is necessary to achieve the objectives of sustainable development.

The main finding of this research focuses on the analysis of the degree of association and determination in the relationship between digital transformation, accounting digitalization, data protection and security, digitalization of sustainability reporting, and the quality of information provided to interested parties. Different perceptions are identified in members of different professional bodies, overall people, and cloud computing users.

The implications of this study can be found both at the level of entities and policymakers. The main measures at the entity level consist of the following: the adoption of cloud technologies—entities should invest in cloud computing solutions to improve the quality and accessibility of the information provided; staff training—continuous training of staff in using new technologies and managing sustainability information is important; ensuring robust technological infrastructure—developing robust technological infrastructure that supports digital transformation is crucial for long-term success. Regarding accounting policy decisions, the need for standards and a minimum level of regulation is mentioned.

The research is based on a subjective assessment of practitioners in the field, under the influence of bias which could affect the measurement process, through over and underestimation, respectively, and the objectivity of the results. In addition, in the context in which the study focused on members of RCFC and other professional bodies (RCCCA, RCFA), they are valid at the level of practitioners in Romania, with limits in generalization. Also, the study only addresses the perspective of accountants and not of the interested parties. Connecting IT systems in an Industry 4.0 network, which gathers operational information to a common database, such as the cloud, enables multiple queries both by people with different professions and roles in the entity and by various external stakeholders with particular interests and cultures in specific social patterns. The issue addressed is complex, requiring in-depth studies both in terms of the particularities of the integration of digital technologies in sustainability accounting and in reporting in the field.

The potential research areas aim at the development of sustainability accounting and reporting, by going beyond the traditional accounting perspective and including stakeholders and by orienting towards the general objective of the entity's sustainable development and aligning with societal objectives. Specific performance indicators will be developed, which must be monitored to assess the achievement of sustainable objectives and to support the stakeholders' decision in the relationship with the entity. Also, an integrative approach of digital technologies is necessary to make sustainability accounting and reporting processes more efficient. Applying Artificial Intelligence of Things (AIoT) in sustainability accounting and reporting can be addressed to increase data management and analysis, human-machine interactions, and the efficiency of IoT operations. Longitudinal and cross-entity studies for the identification and implementation of good practices regarding the use of Industry 4.0 in sustainability accounting and reporting to increase the degree of implementation of the SDGs are required. This involves both conceptual studies and empirical research that could provide valuable insights regarding this transdisciplinary issue.

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Appendix A

Questionnaire

Entity age: ...

Entity type: micro-entity, small entity, medium/large entity, association/foundation, public institution

Digital technology used: cloud computing, big data analysis, artificial intelligence, IoT, blockchain, and others

Member of professional bodies: Romanian Chamber of Fiscal Consultants—RCFC, Romanian Chamber of Certified and Chartered Accountants—RCCCA, Romanian Chamber of Financial Auditors—RCFA, and others.

Table A1. Questionnaire items.

Items	To a Very Small Extent	To a Small Extent	Neutral	To a Great Extent	To a Very Large Extent
Is there reluctance to change involved in using digital technologies?					
Does the organization promote the digitization of services previously provided by other means?					
To what extent does the organization use digital solutions for business problems?					
Do the staff have the necessary skills and abilities to use the technologies?					
Do leadership promote and support the digital transformation?					
Does the organization have the financial resources necessary to implement digital technologies?					
Given the complexity of multi-stakeholder involvement and the need for robust data management systems, does the use of digital technology facilitate the collection and analysis of reliable sustainability data?					
To what extent does the organization ensure the automation of sustainable routine, repetitive, and structured accounting operations?					
To what extent does the organization ensure the automation of non-repetitive sustainable accounting processes?					
To what extent does the organization ensure the digitization of relations with interested parties?					
To what extent does the organization ensure the digital workspace?					
To what extent does the organization ensure digital infrastructure and operations?					
To what extent does the organization provide digital products and services?					
Are digital technologies adapted to accounting and integrated with reporting standards?					
Do digital technologies enable sustainable accounting to be tracked and accessed in real-time?					
To what extent are accounting services outsourced to employees' homes?					
To what extent does the accountant need to have digital expertise to correctly measure, analyze, and interpret financial and non-financial data?					
To what extent does the use of digital technology enable accounting professionals to collaborate with cross-functional teams to set performance goals, monitor progress, and provide assurance on the reliability of sustainability information?					
Does the organization adopt shared data access policies and practices?					
Does the organization adopt data security policies and practices?					
Does the organization adopt policies and practices to restrict the creation and deletion of operational–functional structures?					
Does the organization adopt data protection policies and practices?					
Does the application of digital technologies ensure an improvement in the quality of financial reporting?					
Does the application of digital technologies ensure an improvement in the quality of non-financial reporting?					
Does the application of digital technologies provide stakeholders with real-time access to data?					

Table A1. Cont.

Items	To a Very Small Extent	To a Small Extent	Neutral	To a Great Extent	To a Very Large Extent
Does the application of digital technologies ensure faster adoption of decisions by investors?					
Does the application of digital technologies ensure faster adoption of decisions by employees?					
Does the application of digital technologies ensure faster adoption of decisions by customers?					
Does the application of digital technologies ensure faster adoption of decisions by regulatory bodies?					
Does the application of digital technologies ensure risk management and trust building with stakeholders?					
In the absence of universally accepted reporting standards for ESG metrics, does the use of digital technologies ensure a standardization of reporting practices?					
Can effective communication of sustainability information through the use of digital technology increase transparency, accountability, and stakeholder engagement?					
Does the use of digital technology in sustainability accounting and reporting improve access to capital by attracting responsible investors?					
By using digital technology in accounting and reporting, are sustainability concerns focused on internal processes?					
By using digital technology in accounting and reporting, are sustainability concerns focused on the entire lifecycle of the product/service?					
By using digital technology in sustainability accounting and reporting, can organizations improve operational efficiency, reduce costs, and drive innovation by identifying opportunities for sustainable practices throughout their value chains?					
Does the application of digital technologies in accounting ensure an improvement in the quality of information provided to interested parties?					

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